University of Vavuniya

Second Examination in Information Technology - 2022

First Semester - January/February 2024

IT2122 Software Engineering

Answer Four Questions Only

Time Allowed: Two hours

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1. (a) Describe what is meant by software in your own words. allocation of program [10%]

   (b) State and briefly describe four essential attributes of good software. [20%]

   Reliab. Effic. Mainta. Accepta. Scala.

   (c) Explain how a web-based software system is different from a desktop-based software system. [20%]

   (d) Software Process is a structured set of activities required to develop a software system. List and briefly describe the major software process activities. [20%]

   Planning design E

   (e) State the four major activity regions in the spiral model. [10%]

   (f) Critically examine the truthfulness of each of the following statements about the software development process models:

      i. Incremental development model is an efficient model compared to the waterfall model.

      ii. Spiral model is suitable for safety-critical projects such as nuclear reactor systems, airplane computers and life control systems.

   [20%]

RA
SD
DD
P?
Testing
Deploy
Mainten.

2. Consider the following description of a Family Budget Management System: Since the current world financial crisis began, a lot of economists and politicians are working together to find an effective solution for it. Because of the financial crisis, a lot of families in different parts of the world struggle to manage their living expenses. Software engineers, including you, wish to develop a software system that will help these families to manage their budgets. This system aims to cater to a range of financial management needs for users. It facilitates the categorization of monthly and annual expenses, helping users to understand and organize their financial outflows effectively. Then, users can allocate portions of their income to these expenses using the system. Each financial transaction is stored in the system, which could be useful to provide an overview of their spending habits over time. Moreover, it provides various suggestions on budget management strategies. Additional features are incorporated into this system to manage their budget, credits, bills, and loans as well.

Note: Clearly state any assumptions that you think are necessary for the system.

(a) What elicitation methods will you use to collect the requirements of the Family budget management system? Justify your answers. [15%]

(b) Identify four potential stakeholders of the software system. [10%]

(c) State five functional requirements of the software system. [20%]

(d) Deduce three appropriate non-functional requirements of the software system. (Write quantifiable non-functional requirements with specific metrics.) [15%]

(e) Describe the terms, user requirements and system requirements with the aid of an example from the above software system. [20%]

(f) Requirements specification is the process of writing down the requirements in a requirements document. State and briefly describe four appropriate ways of writing the system requirements specification for the Family budget management system. [20%]

3. Consider an online reservation system for a bus company. The bus company owns several buses. Each bus is identified by its vehicle number plate. In addition, buses have manufacturer, model, colour. Customers and the company staff are the main users of the system. Drivers and conductors are considered company staff. All of them have NIC number, name, address, and phone number. All staff have employee id numbers. In addition to that, a driver has a driving license number as well. These buses travel on different trips. Trips have trip number, starting point, ending point, start time and end time. For each trip, one bus is assigned. One or two drivers are assigned for a trip, but exactly one conductor is assigned for a trip. A bus or a staff can be assigned for one or more trips. Buses contain many seats. Seats have seat numbers. A reservation for a seat for a certain trip can be made by a customer. A reservation has a reference number and a price value. A customer can make multiple reservations if he wants to do so.

(a) Draw a class diagram for the above scenario with proper relationships and multiplicity. Classes should contain attributes that are applicable to them. [50%]

(b) For a trip, seats can be viewed by guest users, but can only be reserved by customers through the website of the bus company. The customer has the option to pay for the reservation through the website and the payment provider will process the payment made by the customer. Through the system, a conductor can verify the reservation made by the customer, before starting the trip. A reservation can be canceled by the customer or the administrator of the bus company if it is made 24 hours before the start time. If a reservation is cancelled, a refund is initiated by the payment provider.

Draw a use case diagram describing the functional requirements of the above system. [20%]

(c) Nuwan is a customer who is going to reserve a seat using the bus reservation system. First, he logs in to the system and searches for trips from Colombo to Vavuniya on 2024.03.01. The system displays available trips that match the search filter. Then, Nuwan selects a trip, and available seats are displayed by the system. Nuwan selects a seat. The system checks the availability of the seat again as it is possible for another customer to book that seat within the timeframe. If the seat is not available, an updated availability list is sent by the system to Nuwan, and he selects another seat. If the seat is available, the system asks for credit/debit card details. Nuwan enters all the details. Credit/debit card details are sent to the payment provider from the system and the payment provider checks whether it is possible to process the payment. If it is not possible because of an error or insufficient balance, a notification is sent to the system by the payment provider. Then, the system asks for the correct credit/debit card details from Nuwan. If the payment is processed successfully, the payment provider notifies the system, and the seat will be reserved by the system. After that, a ticket is sent to Nuwan. Draw a sequence diagram for the sequence of activities in seat reservation. [30%]

4. (a) The class diagram is a static diagram, and it shows a collection of classes, interfaces, associations, collaborations, and constraints.
Explain each of the following concepts of a class diagram with the aid of examples and appropriate notations. These notations should be clearly illustrated in diagrams for each of the following concepts:

   i. Class
   ii. Class Associations and Association Multiplicity
   iii. Aggregation and Composition
   iv. Specialization and Generalization

[40%]

(b) Mentcare is a patient information system, which maintains information about patients suffering from mental health problems and the treatments that they have received. Most mental health patients do not require dedicated hospital treatment but need to attend specialist clinics regularly where they can meet a doctor who has detailed knowledge of their problems. These clinics are run in local medical practices or community centers in every city. It makes use of a centralized database of patient information so that it can be accessed and used from the hospitals and clinics. Key features of the Mentcare system are given below:

- Doctors can create records for patients, edit the information in the system, view patient history, etc.

- The system produces data summaries so that doctors can quickly learn about the key problems and treatments that have been prescribed.
- The system monitors the records of patients that are involved in treatment and issues warning if possible problems are detected.
- The system generates monthly management reports showing the number of patients treated at each clinic, the number of patients who have entered and left the care system, number of patients sectioned, and the drugs prescribed, etc.

i. Explain how the Mentcare system can be broken into its subsystems in the software design process. [15%]

ii. It is essential to ensure that the patient information remains confidential and that only authorized persons should be able to access the system. Describe three design strategies that you would include in Mentcare system design to allow only authorized access. [15%]

iii. Design and draw a suitable architecture style for the Mentcare system. Justify your answer. [15%]

iv. Mention five important design considerations for Graphical User Interface design of the Mentcare system. [15%]

5. (a) Explain why good software design is important when it comes to development. [15%]

(b) List five factors that can influence the architectural design of a mobile application. [15%]

(c) Describe the Data-Flow Architecture style in architectural design with the aid of a diagram. [15%]

(d) State and briefly describe the four main types of software maintenance activities. [20%]

(e) State and briefly describe two important reasons why software maintenance is an essential part of software development. [10%]

(f) Briefly describe five key issues in the software maintenance process. [25%]